

Astra, Opus and the revolved mesher

Three solid profiles

Prepared by Ryan | 8 October 2026

Independent LLM-authored block layouts and deterministic CAD-to-mesh results for three bore-free solids of revolution. The examples vary stepped shoulders, chamfered tips, rounded ends and an exact circular fillet.

Part	Approach	Status	Hexes	Min SJ / doubled	Checks	Time	Usage*
01	GPT 6 Astra	Pass	2,898	0.6917 / 0.6967	1	2.22 m	\$1.01
01	Opus 5.5	Pass	2,464	0.6946 / 0.6996	1	3.02 m	\$0.73
01	Revolved mesher	Pass	8,064	0.6924 / 0.6935	1	0.024 s	No API
02	GPT 6 Astra	Pass	4,500	0.6964 / 0.7006	1	2.28 m	\$1.02
02	Opus 5.5	Pass	3,584	0.6946 / 0.6996	1	4.03 m	\$0.93
02	Revolved mesher	Pass	6,180	0.6855 / 0.6860	1	0.043 s	No API
03	GPT 6 Astra	Pass	1,824	0.5774 / 0.5774	1	2.22 m	\$0.93
03	Opus 5.5	Pass	1,740	0.6428 / 0.6428	1	6.55 m	\$1.34
03	Revolved mesher	Refused	-	- / -	0	0.002 s	No API

*GPT dollars are Standard API-equivalent estimates from cumulative native tokens; Claude dollars are CLI-reported usage. Neither is a subscription invoice. Deterministic time covers compilation; model time covers the author session. Astra uses medium reasoning, Opus high.

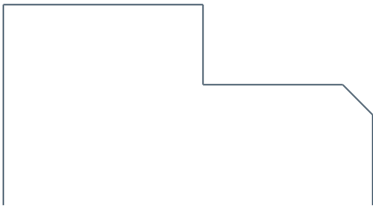
Explicitly synthetic showcase, not a population benchmark. The original CAD and native geometry checks are shared within this experiment. The filleted case records the current automatic-route refusal honestly. No prior or other-model layouts supplied to authors; no expert grades assigned.

Stepped shaft with chamfered nose

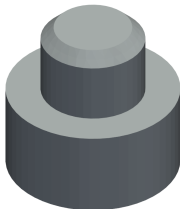
Original CAD

Full solid of revolution.
No holes or internal cavities.
Gray views show input CAD.

MERIDIAN SECTION



ISOMETRIC



Views and quality

Block layout and mesh section are cuts through the actual 3D mesh. SJ colors expose the section interior; both densities are checked.

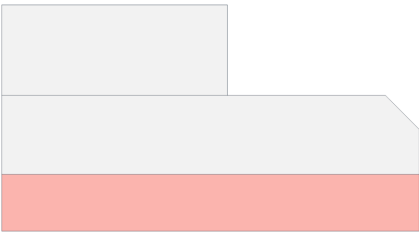
GPT 6 Astra

Geometry pass

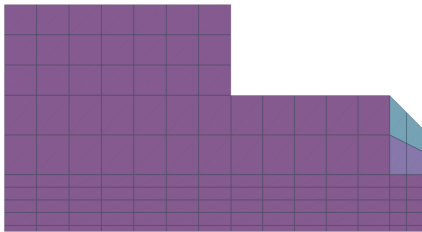
9 seed parts | 2898 hexes
SJ 0.6917 / 2x 0.6967
1 full checks | 2.22 min
Usage \$1.01

Standard API equivalent

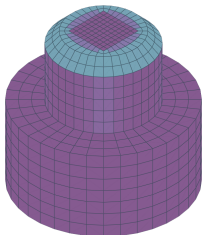
BLOCK SECTIONS



MESH SECTION / SJ



3D MESH / SJ



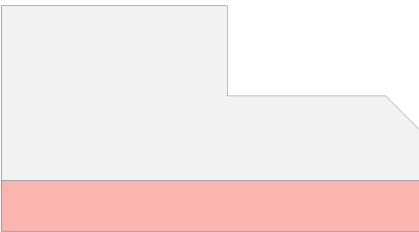
Opus 5.5

Geometry pass

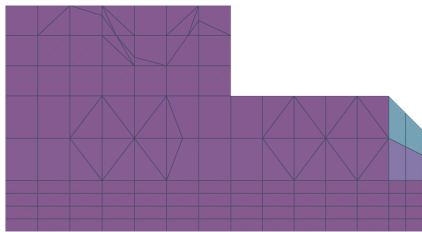
5 seed parts | 2464 hexes
SJ 0.6946 / 2x 0.6996
1 full checks | 3.02 min
Usage \$0.73

Claude CLI usage

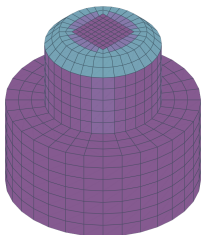
BLOCK SECTIONS



MESH SECTION / SJ



3D MESH / SJ



Revolved mesher

Geometry pass

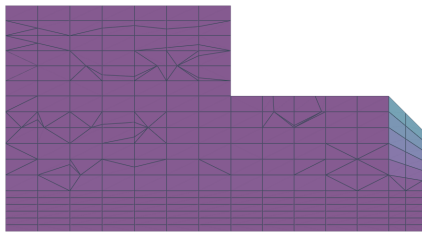
5 seed parts | 8064 hexes
SJ 0.6924 / 2x 0.6935
1 full checks | 0.024 s
Usage no API

local deterministic computation

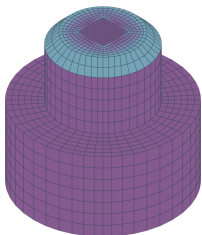
BLOCK SECTIONS



MESH SECTION / SJ



3D MESH / SJ



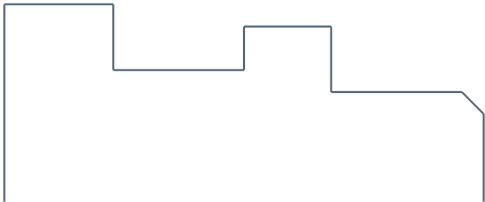
SJ 0.2-1; larger is better. Seed parts may contain several blocks.

Multi-step shaft with chamfered tip

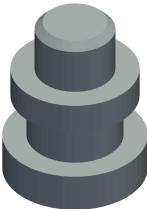
Original CAD

Full solid of revolution.
No holes or internal cavities.
Gray views show input CAD.

MERIDIAN SECTION



ISOMETRIC



Views and quality
Block layout and mesh section are cuts through the actual 3D mesh. SJ colors expose the section interior; both densities are checked.

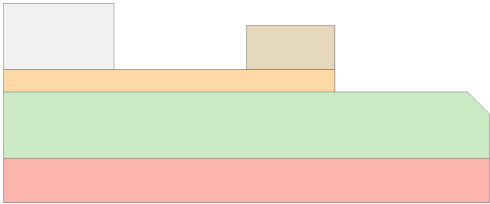
GPT 6 Astra

Geometry pass

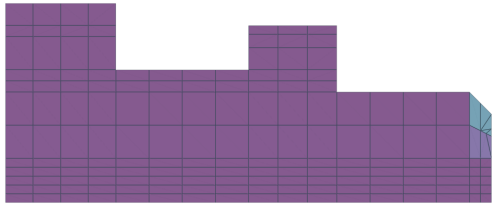
17 seed parts | 4500 hexes
SJ 0.6964 / 2x 0.7006
1 full checks | 2.28 min
Usage \$1.02

Standard API equivalent

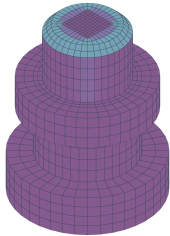
BLOCK SECTIONS



MESH SECTION / SJ



3D MESH / SJ



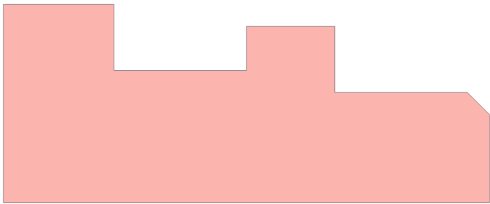
Opus 5.5

Geometry pass

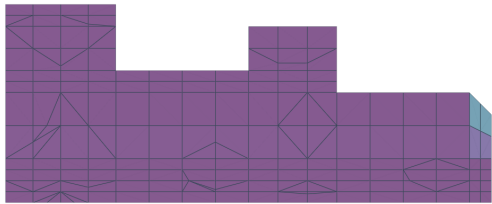
1 seed parts | 3584 hexes
SJ 0.6946 / 2x 0.6996
1 full checks | 4.03 min
Usage \$0.93

Claude CLI usage

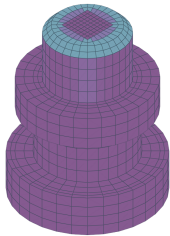
BLOCK SECTIONS



MESH SECTION / SJ



3D MESH / SJ



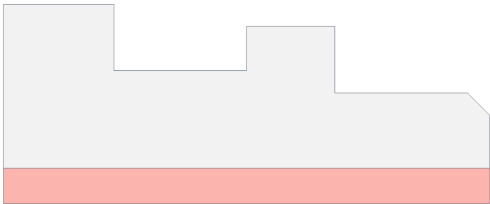
Revolved mesher

Geometry pass

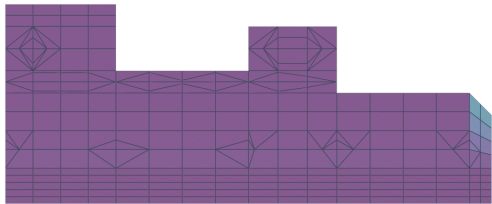
5 seed parts | 6180 hexes
SJ 0.6855 / 2x 0.6860
1 full checks | 0.043 s
Usage no API

local deterministic computation

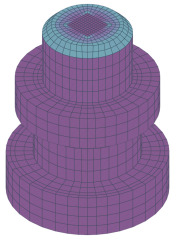
BLOCK SECTIONS



MESH SECTION / SJ



3D MESH / SJ



SJ 0.2-1; larger is better. Seed parts may contain several blocks.

Filletted tapered body with rounded ends

Original CAD

Full solid of revolution.
No holes or internal cavities.
Gray views show input CAD.

MERIDIAN SECTION



ISOMETRIC



Views and quality

Block layout and mesh section are cuts through the actual 3D mesh. SJ colors expose the section interior; both densities are checked.

Fillet detail: Astra represents the tiny fillet with 48/192 boundary quads; Opus spans it (0 attributed quads). Both pass native checks; neither has an eligible sharp edge in this case.

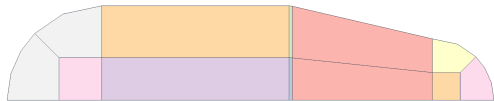
GPT 6 Astra

Geometry pass

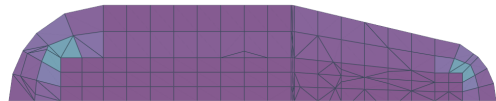
27 seed parts | 1824 hexes
SJ 0.5774 / 2x 0.5774
1 full checks | 2.22 min
Usage \$0.93

Standard API equivalent

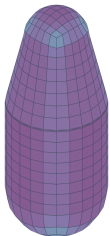
BLOCK SECTIONS



MESH SECTION / SJ



3D MESH / SJ



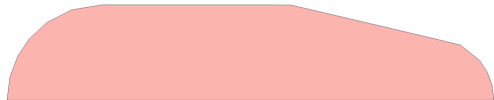
Opus 5.5

Geometry pass

1 seed parts | 1740 hexes
SJ 0.6428 / 2x 0.6428
1 full checks | 6.55 min
Usage \$1.34

Claude CLI usage

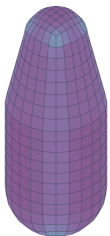
BLOCK SECTIONS



MESH SECTION / SJ



3D MESH / SJ



Revolved mesher

Refused

- seed parts | - hexes
SJ - / 2x -
0 full checks | 0.002 s
Usage no API

local deterministic computation

No layout or mesh emitted.

The automatic route refuses a shallow surface junction. Its two-surface projection construction cannot guarantee convergence here. The CAD and checks were not changed.



SJ 0.2-1; larger is better. Seed parts may contain several blocks.